

AI Interview Platform

Session Cost Analysis Report

Configuration: 30-Question Session | 1-Hour Duration

Date: June 26, 2026

Classification: Internal - Executive Review

Without Prompt Caching

\$0.58

With Prompt Caching

\$0.46

20% cost reduction

1. Executive Summary

This report details the per-session cost of running a 1-hour AI-powered interview with 30 questions. All pricing has been verified against provider pricing pages as of June 26, 2026. Prompt caching on the LLM reduces session cost by 20%.

Service Stack

Function	Service / Model	Provider
Speech-to-Text (STT)	gpt-4o-mini-transcribe	OpenAI
Text-to-Speech (TTS)	Inworld TTS 1.5 Mini	Inworld AI
Question Gen & Evaluation	Claude Haiku 4.5	Anthropic (AWS Bedrock)
AI Answer Detection	Sapling AI Detector API	Sapling
Face Verification	AWS Rekognition CompareFaces	AWS

Without Prompt Caching

\$0.58

With Prompt Caching (Recommended)

\$0.46

Save \$0.12 per session | 20% reduction

2. Cost Breakdown Per Session

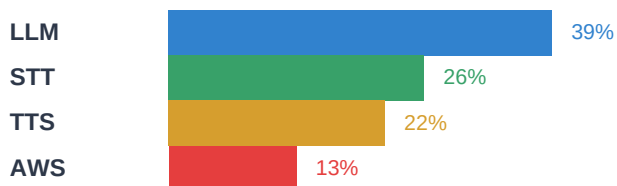
2.1 Without Prompt Caching

Component	Verified Rate	Usage / Session	Cost (USD)
STT - gpt-4o-mini-transcribe	\$0.003 / min	~40 min speech	\$0.120
TTS - Inworld TTS 1.5 Mini	\$25 / 1M chars	~4,000 chars	\$0.100
LLM - Claude Haiku 4.5	\$1.00 / \$5.00 per 1M	~196K in / ~20K out	\$0.296
AI Detection - Sapling API	Free	30 calls	\$0.000
Face Verify - AWS Rekognition	~\$0.001 / image	~63 API calls	\$0.063
TOTAL PER SESSION			\$0.579

2.2 With Prompt Caching (Recommended)

Component	Verified Rate	Usage / Session	Cost (USD)
STT - gpt-4o-mini-transcribe	\$0.003 / min	~40 min speech	\$0.120
TTS - Inworld TTS 1.5 Mini	\$25 / 1M chars	~4,000 chars	\$0.100
LLM - Claude Haiku 4.5	cached rates	~196K in / ~20K out	\$0.180
AI Detection - Sapling API	Free	30 calls	\$0.000
Face Verify - AWS Rekognition	~\$0.001 / image	~63 API calls	\$0.063
TOTAL WITH CACHING			\$0.463

2.3 Cost Distribution (With Prompt Caching)



3. Prompt Caching - How It Saves 20%

Anthropic's prompt caching allows repeated message prefixes to be stored and reused across API calls. In our interview flow, each question-generation call builds on the previous one's input (system prompt + growing conversation history), making it an ideal candidate for caching.

3.1 Caching Rates (Claude Haiku 4.5)

Token Type	Rate (per 1M tokens)	vs Standard
Standard input (no cache)	\$1.00	baseline
Cache write (first occurrence)	\$1.25	+25% (one-time)
Cache read (subsequent hits)	\$0.10	-90% discount
Output (unchanged)	\$5.00	no change

3.2 Why Question Generation Benefits Most

Each question-generation call includes the full conversation history as a prefix. Call N reuses 100% of Call N-1's input from cache, only writing the new Q&A pair (~250 tokens) to cache. By Call 30, ~97% of input tokens are cache reads at 90% discount.

Call #	Total Input	Cache Read (at \$0.10/1M)	Cache Write (at \$1.25/1M)
Call 1	2,000 tokens	0	2,000 tokens
Call 5	3,000 tokens	2,750	250
Call 10	4,250 tokens	4,000	250
Call 20	6,750 tokens	6,500	250
Call 30	9,250 tokens	9,000	250

3.3 Token Breakdown (Question Generation)

Metric	Without Caching	With Caching
Total input tokens	~140,000	~140,000
Tokens billed at standard rate	140,000 at \$1.00/1M	--
Tokens billed as cache reads	--	~131,000 at \$0.10/1M
Tokens billed as cache writes	--	~9,000 at \$1.25/1M
Question gen input cost	\$0.140	\$0.024
Savings		83%

3.4 Full LLM Cost Comparison

LLM Call Type	Without Cache	With Cache	Savings
Question generation (input)	\$0.140	\$0.024	-83%
Expected points (input)	\$0.036	\$0.036	--
AI detection fallback (input)	\$0.007	\$0.007	--
Evaluation + greeting (input)	\$0.013	\$0.013	--
All output tokens	\$0.100	\$0.100	--
TOTAL LLM COST	\$0.296	\$0.180	-39%

Prompt caching reduces LLM cost by 39% (\$0.296 -> \$0.180) with zero impact on output quality. The feature requires no code changes to the question-generation prompt -- only enabling the `cache_control` flag on the system prompt.

4. LLM Token Detail (Claude Haiku 4.5)

Claude Haiku 4.5 is used for question generation, answer evaluation, expected-point extraction, and AI detection fallback. Pricing: \$1.00 per 1M input tokens (or cached rates), \$5.00 per 1M output tokens.

LLM Call Type	Calls	Input Tokens	Output Tokens	Cost (USD)
Greeting generation	1	500	30	\$0.001
Question generation	30	140,000	3,000	\$0.155
Expected points extraction	30	36,000	6,000	\$0.066
AI detection fallback	~12	7,200	600	\$0.010
Final evaluation	1	12,000	10,000	\$0.062
TOTAL (without cache)	~74	~196,000	~20,000	\$0.296
TOTAL (with cache)	~74	~196,000	~20,000	\$0.180

Why does question generation dominate? Each successive question includes the full conversation history. By question 30, the input context contains all 29 prior Q&A pairs (~9,000+ tokens per call). Prompt caching turns this from a cost problem into a cost advantage -- the growing prefix is read from cache at 90% discount.

5. Service-by-Service Detail

5.1 Speech-to-Text - gpt-4o-mini-transcribe

Parameter	Value
Rate	\$0.003 per minute (\$0.18 per hour)
Candidate speaking time	~40 minutes of 60-minute session
How it's used	Each answer is streamed and transcribed in real-time
Per-session cost	\$0.120

Source: openai.com/api/pricing (verified June 2026)

5.2 Text-to-Speech - Inworld TTS 1.5 Mini

Parameter	Value
On-demand rate	\$25 per 1M characters (\$0.025 per 1K chars)
Scale rate (Enterprise)	\$5 per 1M characters (\$0.005 per 1K chars)
Characters per session	~4,000 (1 greeting + 30 questions streamed)
Per-session cost (on-demand)	\$0.100
Per-session cost (at scale)	\$0.020

Source: inworld.ai/pricing (verified June 2026)

5.3 AI Detection - Sapling API

Parameter	Value
Rate	Free (AI Detector API is free of charge)
How it's used	Each answer (first 2,000 chars) checked for AI content
Fallback	If score is ambiguous (0.4-0.7), LLM verifies (cost in LLM)
Per-session cost	\$0.000

Source: sapling.ai/docs/api/pricing (verified June 2026). High-volume may require contacting Sapling.

5.4 Face Verification - AWS Rekognition

Parameter	Value
APIs used	DetectFaces (registration) + CompareFaces (periodic)
Rate	~\$0.001 per image (first 1M images/month)
Calls per session	~3 detect + ~60 compare (every 60s) = ~63 total

Per-session cost	\$0.063
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Source: aws.amazon.com/rekognition/pricing (verified June 2026). Free tier: 1,000 images/month for first 12 months.

6. Cost at Scale

6.1 Monthly Projections (With Prompt Caching)

Monthly Volume	STT	TTS	LLM (cached)	AI Detect	AWS	Total
100 interviews	\$12	\$10	\$18	\$0	\$6	\$46
500 interviews	\$60	\$50	\$90	\$0	\$32	\$232
1,000 interviews	\$120	\$100	\$180	\$0	\$63	\$463
5,000 interviews	\$600	\$500	\$900	\$0	\$315	\$2,315
10,000 interviews	\$1,200	\$1,000	\$1,800	\$0	\$630	\$4,630

6.2 Savings Summary by Optimization

Configuration	Per Session	1,000/mo	10,000/mo	Savings
Baseline (no optimizations)	\$0.579	\$579	\$5,790	--
+ Prompt caching	\$0.463	\$463	\$4,630	20%
+ Prompt cache + scale TTS	\$0.383	\$383	\$3,830	34%
+ All optimizations*	\$0.319	\$319	\$3,190	45%

* All optimizations = prompt caching + enterprise TTS (\$5/1M) + batch processing for evaluation (50% LLM discount on non-realtime calls).

With all optimizations at 10,000 interviews/month: \$3,190/month total. That is \$0.32 per interview vs \$15-50+ for human screening -- a 47x-156x cost reduction.

7. Alternative LLM Cost Comparison

The LLM is the largest cost lever. Below is a comparison across available models (same ~196K input / ~20K output token profile). Prompt caching benefit shown for Claude models that support it.

Model	Input/1M	Output/1M	LLM Cost	With Cache	Session Total
Amazon Nova Micro	\$0.035	\$0.14	\$0.010	N/A	\$0.293
Amazon Nova Lite	\$0.06	\$0.24	\$0.017	N/A	\$0.300
GPT-4o-mini	\$0.15	\$0.60	\$0.041	N/A	\$0.324
Grok 4.1 Fast	\$0.20	\$0.50	\$0.049	N/A	\$0.332
Haiku 4.5 [Selected]	\$1.00	\$5.00	\$0.296	\$0.180	\$0.463
Claude Sonnet 4.6	\$3.00	\$15.00	\$0.888	\$0.536	\$0.819
Claude Opus 4.6	\$15.00	\$75.00	\$4.440	\$2.680	\$2.963

Claude Haiku 4.5 with prompt caching (\$0.46/session) offers the best quality-to-cost balance. It is 7x cheaper than Sonnet while maintaining strong reasoning for contextual follow-up questions and detailed evaluations.

8. Assumptions & Notes

Session Parameters

- Session duration: 1 hour (60 minutes)
- Number of questions: 30
- Average candidate speaking time: ~40 minutes (67% of session)
- Average answer length: ~1 min speech (~150 words, ~750 characters)
- Face verification interval: Every 60 seconds
- AI detection: Runs on every candidate answer

Prompt Caching Assumptions

- Cache TTL: 5 minutes (Anthropic default; sufficient within a session)
- Cache hit rate on question gen: ~94% of input tokens are cache reads
- Only question-generation calls benefit (growing prefix pattern)
- Other LLM calls have unique content per call, minimal cache benefit
- Requires `cache_control` flag on system/context messages

Token Estimation Method

- Claude models: ~3.5 characters per token
- Conversation history grows linearly with each question
- Question 1 input: ~2,000 tokens; Question 30 input: ~9,250 tokens

Pricing Sources (Verified June 26, 2026)

Service	Source
OpenAI STT	openai.com/api/pricing
Inworld TTS	inworld.ai/pricing
Claude Haiku 4.5	platform.claude.com/docs/en/about-claude/pricing
Sapling AI Detection	sapling.ai/docs/api/pricing
AWS Rekognition	aws.amazon.com/rekognition/pricing

What's NOT Included

- Server hosting / compute costs (EC2, database)
- Network bandwidth / data transfer
- Resume parsing (one-time per candidate, negligible)
- Storage costs (session data, recordings)

Bottom Line:

\$0.46 per interview

with prompt caching | vs \$15-\$50+ for human screening